Ram Cauchy Moments for Critical Honeycomb Strip Mass: Exact Representations, Endpoint Certificates, and a Stable Hard-Edge Extension

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
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Abstract

The assigned OpenAI manuscript identifies a honeycomb-strip arch moment with an expectation in a positive Cauchy-interaction ensemble and claims uniform crossing and arch-moment powers $N^{-1/4}$ and $N^{3/4}$. We preserve its path normalization and separate the integrable-model identification from the positive-integral analysis. For the latter we derive a finite Ram moment system, a Laplace-feature representation, and a primal–dual error certificate with explicit singular-endpoint and Laplace-tail budgets. The actual response matrix has a rank-one symmetric part and is not a positive-definite Gram matrix; its invertibility comes from Cauchy determinant positivity, not a coercivity substitution. We prove that a class of positive weights with hard-edge power x^{-p}, $0 < p < 1/2$, and two specified logarithmic-slope bounds has integrated ratio observable of order N^{3p}. This includes bounded C^1 logarithmic perturbations with bounded derivative of the manuscript's weight and preserves its $3/4$ integral exponent. The extension is for the positive ensemble; no new lattice realization, self-avoiding-walk scaling limit, or uniform-in-height matrix conditioning is asserted. The integrability threshold, exact weight sensitivities, and the extra hypotheses needed to recover a bridge exponent are established explicitly. Contribution assessment: C3/5, a notable positive-integral stability extension and source-specific certification framework, without an independently certified new global lattice theorem or demonstrated computational speedup.

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1 Question, sources, and evidential scope

The target is OpenAI, *Critical strip-crossing mass on the honeycomb lattice*, September 26, 2026, manuscript M463 in family 237 [1]. Its local snapshot is pinned to repository commit `adc7f124 1b42e322a6451854ab7e4b4c146bf78a`. The project-level upstream check on October 8 compared selected public text sources against commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`; it found no change to this selected manuscript. This is a source-identity check, not a proof audit or a check of every transitive mathematical dependency.

The report asks which exact problem can be reformulated in Ram coordinates, what this changes mathematically, and where global claims still require the original argument. Its strongest extension concerns the positive integral after the manuscript's lattice-to-integral identification. This distinction is essential: a theorem for a perturbed measure is not automatically a theorem for a perturbed honeycomb walk.

1.1 Current primary literature

Duminil-Copin and Smirnov established the honeycomb connective constant $\sqrt{2+\sqrt{2}}$ using a parafermionic observable [2]. Krachun and Panagiotis established quantitative sub-ballisticity and polynomial decay of critical strip bridge mass [3]. Their work does not by itself supply the precise exponent claimed in M463. Cauchy biorthogonal polynomials and Bures/Cauchy ensemble identities predate the assigned manuscript [4, 5]. In particular, determinant positivity, biorthogonality, and the use of Cauchy kernels are not inventions of Ram Calculus. Public primary-source metadata and relevant statements were checked on October 8, 2026; this was not an exhaustive priority search or a complete audit of those older papers.

The repository's family-237 scope page describes a Lean formalization of the strip result. The linked comparator file, as inspected on the same date, ends its theorem `critical_strip_mass` with `sorry` [6]. That inspected file is therefore a statement with an admitted proof, not an independently checked proof of the headline theorem. No Lean build or hidden proof package is credited here.

1.2 Direct and conditional statements

The calculations below fall into three classes. First, the geometry, transfer identities, and identification of an arch coefficient are the assigned source's claims, with their actual proof obligations retained. Second, Sections 4–7 prove exact finite-dimensional and positive-integral statements directly. Third, the hard-edge argument in Section 8 uses the explicit classical Bures–Laguerre normalizer and the accompanying coordinate bounds, for which the derivation is given here. Recovering a physical bridge estimate additionally needs the source's monotonicity and increment comparison. Finite diagnostics support implementation checks; none replaces an all-N or all-weight proof.

2 The original problem and its proof architecture

The lattice is dual to the equilateral triangular tiling. Horizontal band boundaries are separated by $\sqrt{3}/2$. Ports are boundary-edge midpoints. A path uses the two endpoint half-edges, visits each dual vertex at most once, and has no other boundary contact. Its length counts visited dual vertices, not ports. Set

$$\rho = (2+\sqrt{2})^{-1/2}, \quad \lambda = \pi/8, \quad c = \cos(3\pi/8), \quad C = \cos(\pi/8), \quad d = \cos(\pi/4), \quad \eta = 2d - 1 > 0. \tag{1}$$

For a strip of $N \geq 1$ bands, a fixed bottom source at port zero, and bottom ports with spacing one, the source defines

$$K_N(k) = \sum_{\gamma:0\to k} \rho^{|\gamma|}, \qquad \mathcal{A}_N = \sum_{k\neq 0} K_N(k), \qquad m_N = \sum_{k\geq 1} kK_N(k). \tag{2}$$

The bridge mass $\mathcal{B}_N$ sums paths from the same source to all top ports, which belong to $N/2 + \mathbb{Z}$. These are unnormalized partition sums. The theorem claimed in the source is

$$c\mathcal{A}_N + \mathcal{B}_N = 1, \quad m_{N+1} - m_N \asymp \mathcal{B}_N, \quad m_N \asymp N^{3/4}, \quad \mathcal{B}_N \asymp N^{-1/4}, \tag{3}$$

with finite sums, constants independent of N, and decreasing $\mathcal{B}_N$. An exponent for this height-indexed, all-length moment does not prove the fixed-length displacement or conformal-limit conjecture for self-avoiding walk.

Source stage	Necessary mathematical content	Ram use and limitation
Flux and finite transfers	Boundary cancellation; subcritical transfer on the nonvacuum/source sectors at fixed height; correct link-state gluing.	Exact finite response coordinates and residuals are available only after the physical state space and its inverse exist. Positivity of path weights does not certify arbitrary transfer inverses.
Polynomial vacuum	Pfaffian reductions, stationarity, and an actual degree bound before interpolation.	Moment or jet coordinates can store the same finite polynomial. Finite samples without the degree bound are insufficient.
Arch coefficient	Each arch is counted once for every intervening cut; the source resolvent and reflected gluing produce the displacement moment.	The same bilinear observable can be evaluated in any invertible bank. Missing reflected source equations or a changed pairing would change the observable.
Positive integral	Coalescing parameters, missing harmonics, Cauchy bimoments, and a strictly positive determinant.	This is the concrete target of the exact Ram system below. It is biorthogonal, not an ordinary symmetric moment Gram problem.
Hard edge	Positive association, explicit normalizers, small-coordinate tails and inverse moments.	We retain the positive law and prove a slope-controlled class of weight extensions. No fitted finite-N power law is used as proof.
Bridge recovery	Decreasing bridge mass and a two-sided increment comparison.	A quantitative sequence lemma identifies the additional gates; a general weight perturbation does not inherit them.

2.1 Exact object isolated from the source

The source's Section 5 gives the density

$$\omega(x) = \tfrac{1}{2}\sqrt{\sqrt{(1+x)/(2x)} - 1}, \qquad 0 < x < 1, \tag{4}$$

and the law on $0 < y_1 < \cdots < y_N < 1$ proportional to

$$W_N(y) = \prod_i \omega(y_i) \prod_{i<j} \frac{(y_j - y_i)^2}{y_i + y_j}. \tag{5}$$

Its claimed exact physical identity is

$$m_N = \frac{\eta}{\pi} \int_0^1 \left(1 - \mathbb{E} \prod_{i=1}^{N} \frac{y_i - x}{y_i + x}\right) \frac{\omega(x)}{x}\, \mathrm{d}x. \tag{6}$$

The proof uses $h = b^2(1-b^2)$ and the missing-fourth-harmonic decomposition $Q = \mathcal{P}(h) + b\mathcal{J}(h) + b^2\mathcal{E}(h)$. The prescribed zero multiplicities at $b = C, c$ become decay conditions for Cauchy transforms. Their moment equations are exactly those in Section 4. In particular, the nonsingular finite system controls coalescence; one cannot simply assume a nonzero Pfaffian denominator throughout the limiting operation.

For the later proof we use only the positive-integral right side and the stated normalization. No general Ram inversion theorem independently establishes the preceding transfer, degree, or lattice-identification steps.

2.2 Bounded checks of the preceding source steps

An independent read of the source's complete flux-transfer, vacuum, arch-formula, and local-calculation modules found no demonstrated mathematical error in those assigned modules. This is not a certification of the entire theorem. The audit checked the column convention $(\text{south}, \text{west}) \mapsto (\text{east}, \text{north})$ with its shifted local operator, the fixed-height nature of transfer summability, and the degree bounds used before interpolation. In the vacuum proof, the stationarity divisor only saturates the Laurent-degree bound; odd occupied-slot parity supplies the remaining vanishing argument. In the arch proof the reflected source equation is projected explicitly, rather than assuming a splitter is an isometry. The $4N - 3$ generic interpolation roots exceed degree $2N$ only after the separately checked initial cases are supplied.

The stepped-strip increment argument was also traced: noncrossing pairings and exclusion of closed components enforce the order of interface visits, while its return-series ratio is $\rho^2 < 1$ uniformly in height. Exact finite tests reproduced the displayed splitting and unitarity identities, moving-box cancellations, the width-one moment, and four width-two resolvent/coefficient fixtures. A finite enumeration of the step pairings through 15 visits supports the implementation but does not prove the all-size assertion. Arbitrary-height transfers were not generated exhaustively; generic symbolic width-two attempts that were stopped as too slow are not counted as completed checks.

3 Relevant Ram results and the exact translation

Three actual corpus passages are directly applicable. Foundation report TR-rev-1-old-tr117, Section "Basic Ram Transform and Ram Localization," equations `eq:RT`, `eq:RLT`, `eq:IRLT`, proves equality

of a two-point integral and its displacement-coordinate form [7]. Its inverse localization is an inverse change of kernel coordinates, not an inverse of the integral operator. Book volume 1, revised Chapter 3, `thm:ram-local`, gives a finite Taylor-moment formula with an absolute-moment remainder; `thm:degree-closure` separately specifies when a polynomial space is preserved [8]. Foundation TR-rev-1-old-tr250, equation `eq:moments` and "A concrete polynomial Ram construction," distinguishes invertible finite moments from infinite-dimensional recovery [9]. These passages were read, including their hypotheses and proofs. Project-wide corpus maps supplied routing, not substitute evidence.

Let w be a positive integrable weight on $(0,1)$ and set

$$(T_w f)(x) = \int_0^1 \frac{w(y)f(y)}{x+y}\,\mathrm{d}y, \qquad x > 0. \tag{7}$$

In output-centered Ram coordinates $u = x - y$ this is

$$T_w f(x) = \int_{x-1}^{x} \frac{w(x-u)}{2x-u} f(x-u)\,\mathrm{d}u. \tag{8}$$

The moving interval and the denominator must both be retained. For a polynomial P of degree at most $N-1$, define

$$\mu_r(x) = \int_0^1 \frac{(x-y)^r w(y)}{x+y}\,\mathrm{d}y.$$

Then the finite formula is exact:

$$T_w P(x) = \sum_{r=0}^{N-1} \frac{(-1)^r}{r!}\mu_r(x) P^{(r)}(x). \tag{9}$$

This follows by substituting the polynomial Taylor identity, with no infinite-series interchange. For $f \in C^N[0,1]$ and $0 < x \le 1$, the error is bounded by $\left\|f^{(N)}\right\|_\infty \int_0^1 |x-y|^N w(y)/(x+y)\,\mathrm{d}y/N!$. All these integrals are finite for fixed $x > 0$; a bound uniform as $x \downarrow 0$ is a separate question. The coefficients μ_r are not generally polynomials in x. Thus exactness on polynomial inputs does not justify a nilpotent finite inverse for T_w.

4 An exact Cauchy moment system

Assume for now

$$0 < w(x) \le Ax^{-p} \quad (0 < x < 1), \qquad 0 < p < 1/2, \tag{10}$$

with positivity almost everywhere. Define the ordered ensemble (5) with ω replaced by w, its expectation $\mathbb{E}_w$, and

$$G_N(x) = \mathbb{E}_w \prod_i \frac{y_i - x}{y_i + x}, \qquad J_N(w) = \int_0^1 (1 - G_N(x))\frac{w(x)}{x}\,\mathrm{d}x. \tag{11}$$

Its normalizer is finite and positive: every pair kernel is at most one, and the weight is integrable. Put, for $0 \le j, k < N$,

$$u_j = \int_0^1 x^j w(x)\,\mathrm{d}x,$$

$$H_{jk} = \int_0^1 \int_0^1 \frac{x^j y^k}{x+y} w(x)w(y)\,\mathrm{d}x\,\mathrm{d}y, \tag{12}$$

$$M_{jk} = \int_0^1 \int_0^1 \frac{x^{j+1} y^k}{x+y} w(x)w(y)\,\mathrm{d}x\,\mathrm{d}y, \qquad h_k = H_{0k}. \tag{13}$$

The H integrals are finite because

$$\int_0^\infty \frac{y^{-p}}{x+y}\,\mathrm{d}y = \frac{\pi}{\sin(\pi p)}x^{-p}, \qquad \int_0^1 x^{-2p}\,\mathrm{d}x < \infty. \tag{14}$$

The matrix M needs only $p<1$, but the observable and H require the stronger condition used here.

Theorem 4.1 (Finite response and integrated observable)**.** *The matrix M is invertible and* $\det M > 0$*. If $a = M^{-1}u$ and $P_N(y) = \sum_{k=0}^{N-1} a_k y^k$, then*

$$G_N(x) = 1 - xT_w P_N(x), \qquad J_N(w) = h^{\mathsf{T}} a. \tag{15}$$

For $w = \omega$, the assigned source identity gives $m_N = (\eta/\pi)h^{\mathsf{T}}M^{-1}u$.

Proof. Antisymmetrize the determinant integral in both variable families. On their ordered domains, the integrand is the product of two positive Vandermondes, $\prod_i x_i$, the positive weights, and

$$\det[(x_i+y_j)^{-1}] = \frac{\prod_{i<j}(x_j-x_i)(y_j-y_i)}{\prod_{i,j}(x_i+y_j)} > 0.$$

It is positive on a set of positive measure. Expansion of the finite determinant is integrable by the finite entry bounds, so $\det M > 0$. This is the classical Cauchy-bimoment argument, not a symmetric-Gram argument.

For distinct coordinates, partial fractions give

$$\prod_i \frac{y_i - x}{y_i + x} = 1 - x\sum_i \frac{2\prod_{j\neq i}(y_j+y_i)/(y_j-y_i)}{x+y_i}.$$

Multiplication by the density cancels all apparent collision poles: one Vandermonde factor remains and the other factors are bounded polynomials or bounded pair kernels. Integrating the other $N-1$ coordinates produces $G_N(x) = 1 - xT_wP(x)$ for a polynomial of degree at most $N-1$. Every term is absolutely integrable after cancellation. Symmetrizing in $x, y_1, \dots, y_N$ gives

$$\int_0^1 x^j G_N(x) w(x)\,\mathrm{d}x = 0 \quad (0 \le j < N),$$

because $\sum_{r=0}^{N} z_r^j / \prod_{s\neq r}(z_s - z_r) = 0$ for $j < N$. The product ratio is bounded in absolute value by one, so this symmetrization is legitimate. Inserting the Cauchy representation gives $Ma = u$, whose unique solution determines P. Finally, P is bounded on $[0,1]$, and (14) justifies Fubini for $\int w(x)T_wP(x)\,\mathrm{d}x$. This proves the second identity. The last assertion uses (6), not the matrix algebra alone. □

4.1 The matrix is not a coercive Gram matrix

Proposition 4.2 (A rank-one symmetric part)**.** *The exact response obeys*

$$M + M^{\mathsf{T}} = uu^{\mathsf{T}}. \tag{16}$$

For $N > 1$ its symmetric part is singular, although M is invertible. Consequently a positive lower bound for $v^{\mathsf{T}}Mv/\|v\|_2^2$ cannot certify its inverse.

Proof. Exchange x, y in M_{kj} and add the two integrands. Their numerator becomes $x^j y^k(x+y)$, giving $u_j u_k$. For any nonzero $v \perp u$, $v^{\mathsf{T}}Mv = (v^{\mathsf{T}}u)^2/2 = 0$. Such v exists when $N > 1$. This does not imply $Mv = 0$; the antisymmetric part is essential. □

The correct finite stability quantity is an inf–sup or singular-value bound for the actual response. Invertible changes to Ram moments preserve the system if both input and output coordinates are transformed, but changing M into its symmetric part destroys invertibility for $N > 1$. Normal equations are possible but square the Euclidean condition number; they do not supply missing precision for free.

5 Laplace features and a physical positive metric

Define $F_j(t) = \int_0^1 x^j e^{-tx} w(x)\,\mathrm{d}x$ for $t \geq 0$. The elementary positive identity $(x+y)^{-1} = \int_0^\infty e^{-t(x+y)}\,\mathrm{d}t$ gives

$$H_{jk} = \int_0^\infty F_j(t)F_k(t)\,\mathrm{d}t, \qquad M_{jk} = \int_0^\infty [-F_j'(t)]F_k(t)\,\mathrm{d}t. \tag{17}$$

Tonelli applies first to each nonnegative monomial entry; finite linear combinations follow by absolute integrability. This is a Laplace-feature realization of the same Cauchy kernel, compatible with exponential Ram moments, not a new Laplace-transform theorem.

Proposition 5.1 (Positive metric and boundary identity)**.** *H is positive definite. In the feature realization, integration by parts gives* (16)*, with its entire symmetric part coming from $t = 0$, not from a positive bulk quadratic form.*

Proof. If $v^{\mathsf{T}} H v = 0$, the continuous function $\sum_j v_j F_j(t)$ vanishes identically. Differentiation at $t = 0$ shows that $\int x^r P(x) w(x)\,\mathrm{d}x = 0$ for every $r \geq 0$, where $P(x) = \sum_j v_j x^j$. Taking the linear combination corresponding to P gives $\int P^2 w = 0$, hence $P = 0$ and $v = 0$. Integration of $-(F_j F_k)'$ gives $F_j(0)F_k(0) = u_j u_k$, since both transforms tend to zero at infinity. The derivative integrals are absolutely integrable by (13). □

An H-orthonormal basis provides a physical metric on this finite polynomial-feature space. It does not turn the original response into a self-adjoint positive operator. The finite span, the measure, and the boundary term must all remain unchanged in any coordinate comparison.

Proposition 5.2 (Explicit Laplace tails)**.** *Under* (10)*, for $T > 0$,*

$$\int_T^\infty F_j F_k\,\mathrm{d}t \leq \frac{A^2\Gamma(j+1-p)\Gamma(k+1-p)}{j+k+1-2p} T^{-(j+k+1-2p)}, \tag{18}$$

$$\int_T^\infty (-F_j')F_k\,\mathrm{d}t \leq \frac{A^2\Gamma(j+2-p)\Gamma(k+1-p)}{j+k+2-2p} T^{-(j+k+2-2p)}. \tag{19}$$

Proof. Enlarge the x integration to $(0,\infty)$ to obtain $F_j(t) \leq A\Gamma(j+1-p)t^{-(j+1-p)}$ and the analogous derivative bound. Integrating their products gives the formulas. All exponents at the denominator are positive under $p < 1/2$. □

Finite exponential quadrature is therefore accompanied by a separate infinite-t tail, not declared exact because a finite feature bank fits its retained moments. No decay rate of a matrix condition number in N follows from these entrywise estimates.

6 Endpoint-safe numerical and residual certificates

Let $u^\delta, M^\delta, h^\delta$ use $[\delta, 1]$ in every integral. The omitted portions can be bounded before applying an interior quadrature method.

Proposition 6.1 (Endpoint budgets)**.** *For every entry,* $0 < \delta < 1$*,*

$$\begin{aligned} |u_j - u_j^\delta| &\le \frac{A}{1-p}\delta^{1-p} =: e_u, \\ |M_{jk} - M_{jk}^\delta| &\le \frac{2A^2}{(1-p)^2}\delta^{1-p} =: e_M, \\ |h_k - h_k^\delta| &\le \frac{2A^2\pi}{(1-2p)\sin(\pi p)}\delta^{1-2p} =: e_h. \end{aligned} \tag{20}$$

Thus $\left\|u - u^\delta\right\|_2 \le \sqrt{N}e_u$*,* $\left\|M - M^\delta\right\|_2 \le Ne_M$*, and* $\left\|h - h^\delta\right\|_2 \le \sqrt{N}e_h$*.*

Proof. The first bound integrates the envelope. For M, use $x/(x+y) \le 1$ and $x^j, y^k \le 1$, and bound the two omitted strips separately; their overlap may be counted twice. For h, bound the two strips by (14), then integrate x^{-2p} on $(0, \delta)$. The vector and matrix norm estimates follow from entrywise bounds and the Frobenius norm. □

For the manuscript weight one may take $p = 1/4, A = 1/2$ directly from (4). The observable's endpoint budget is of order $\delta^{1/2}$, slower than the moment-response budget of order $\delta^{3/4}$. This explains why a small residual of an interior matrix alone is not a complete observable certificate.

Suppose interior quadrature and endpoint/tail bounds give

$$\left\|M - \widehat{M}\right\|_2 \le \epsilon_M, \quad \|u - \widehat{u}\|_2 \le \epsilon_u, \quad \left\|h - \widehat{h}\right\|_2 \le \epsilon_h.$$

Let s_0 be a *certified* lower bound for $\sigma_{\min}(\widehat{M})$ and assume $\beta = s_0 - \epsilon_M > 0$. Ordinary floating-point singular values are not certified bounds without rounding/error control. For arbitrary approximate primal and dual vectors $\widehat{a}, \widehat{z}$, set

$$\begin{aligned} r_0 &= \widehat{u} - \widehat{M}\widehat{a}, & s_0^{\mathrm{res}} &= \widehat{h} - \widehat{M}^{\mathsf{T}}\widehat{z}, \\ R &= \|r_0\|_2 + \epsilon_u + \epsilon_M \|\widehat{a}\|_2, & S &= \|s_0^{\mathrm{res}}\|_2 + \epsilon_h + \epsilon_M \|\widehat{z}\|_2. \end{aligned} \tag{21}$$

Theorem 6.2 (Primal–dual observable certificate)**.** *With* $\widehat{J} = \widehat{h}^{\mathsf{T}}\widehat{a} + \widehat{z}^{\mathsf{T}}r_0$*,*

$$|J_N(w) - \widehat{J}| \le \epsilon_h \|\widehat{a}\|_2 + \|\widehat{z}\|_2 (\epsilon_u + \epsilon_M \|\widehat{a}\|_2) + \frac{RS}{\beta}. \tag{22}$$

Also $\|a - \widehat{a}\|_2 \le R/\beta$*.*

Proof. The singular-value perturbation inequality gives $\|M^{-1}\|_2 \le 1/\beta$. Let $r = u - M\widehat{a}$ and $s = h - M^{\mathsf{T}}\widehat{z}$. Then

$$J_N - h^{\mathsf{T}}\widehat{a} - \widehat{z}^{\mathsf{T}}r = s^{\mathsf{T}}M^{-1}r.$$

The residual bounds give RS/β; the difference between the two computed linear terms and their exact counterparts gives the remaining terms in (22). The primal bound follows from $a - \widehat{a} = M^{-1}r$. □

This standard adjoint-residual mechanism is useful here because it certifies the actual arch functional, including the singular measure, without requiring each coefficient to be extremely accurate. Its usefulness remains conditional on computable enclosures and a nonzero response margin. Floating-point evaluation of the residuals, norm bounds, and corrected output must itself be enclosed; any output-rounding error is added to the displayed bound. A finite computation at selected heights does not prove a uniform $N^{-1/4}$ law.

7 Exact sensitivity and finite weight perturbations

Let $w_t = e^{tV}w$ with bounded real V and t in a compact interval. The envelope persists with a changed constant, so all finite integrals and their derivatives are dominated. Write $M(t)a(t) = u(t)$ and $J(t) = h(t)^{\mathsf{T}}a(t)$. Then

$$a' = M^{-1}(u' - M'a), \qquad J' = h'^{\mathsf{T}}a + z^{\mathsf{T}}(u' - M'a), \quad M^{\mathsf{T}}z = h. \tag{23}$$

Here $u_j' = \int x^j V(x) w_t(x)\,\mathrm{d}x$, and M', h' insert the factor $V(x) + V(y)$ into their double integrals. These formulas preserve the two-sided weight dependence. Differentiating just the right-hand side of the finite system would omit a generally nonzero term.

Proposition 7.1 (Normalization-aware perturbation)**.** *For bounded V, with $v_- \le V \le v_+$ almost everywhere and $\Delta = v_+ - v_-$,*

$$e^{v_- - N\Delta} J_N(w) \le J_N(e^V w) \le e^{v_+ + N\Delta} J_N(w). \tag{24}$$

The derivative obeys

$$|J'(t)| \le (\|V\|_\infty + N \operatorname{osc} V) J(t). \tag{25}$$

Proof. The ensemble likelihood ratio is $e^{\sum_i V(y_i)} / \mathbb{E}_w e^{\sum_i V(y_i)}$, between $e^{-N\Delta}$ and $e^{N\Delta}$. The integrand $1 - \prod_i (y_i - x)/(y_i + x)$ is nonnegative. Multiply these bounds and the one-variable factor $e^{V(x)}$, then integrate. For the derivative, differentiation of the normalized expectation gives a covariance with $\sum_i V(y_i)$, as well as the derivative of the outside weight. For nonnegative F and bounded S, $|\operatorname{Cov}(F, S)| \le \operatorname{osc}(S)\mathbb{E}F$. Apply this pointwise in x and integrate. □

For constant $V = v$, the law does not change and $J_N(e^v w) = e^v J_N(w)$, as both formulas correctly indicate. For a nonconstant fixed tilt, (24) is exponentially weak in N. A uniform exponent stability result therefore needs extra structure; it cannot be inferred from finite-dimensional continuity. The next section supplies such structure through order and logarithmic slopes.

8 A hard-edge extension with uniform height constants

Definition 8.1 (Admissible hard-edge weight)**.** Fix $0 < p < 1/2$. A positive C^1 weight w on $(0, 1)$ is admissible if constants $A, a_*, D, E < \infty$, with $A, a_* > 0$, satisfy

$$\begin{aligned} w(x) &\le A x^{-p} && (0 < x < 1), \\ w(x) &\ge a_* x^{-p} && (0 < x \le 1/2), \\ (\log(w(x)x^{1/2}))' &\ge -D && (0 < x \le 1/2), \\ \left(\log w\left(\frac{2z}{1+z}\right)\right)' &\le E && (0 < z < 1). \end{aligned} \tag{26}$$

There is no monotonicity requirement on w itself and no bounded derivative requirement near the endpoints beyond these one-sided bounds.

Theorem 8.2 (Uniform power for an extended ensemble)**.** *For every admissible weight, there are $0 < c_w \le C_w < \infty$ such that*

$$c_w N^{3p} \le J_N(w) \le C_w N^{3p} \qquad (N \ge 1). \tag{27}$$

The constants may depend on the weight-class parameters and p, but not on N. This theorem concerns (11); it does not assert a corresponding self-avoiding-path model for every w.

We give the ingredients and proof in detail to make clear what is extended and what is inherited from classical positive-measure analysis.

8.1 Pointwise reduction to inverse statistics

For fixed ordered y put $R_y(x) = \prod_i (y_i - x)/(y_i + x)$ and $S_y = \sum_i y_i^{-1} \geq 1$. Each factor lies in $[-1, 1]$; telescoping therefore gives

$$0 \leq 1 - R_y(x) \leq 2 \min(1, xS_y). \tag{28}$$

If $y_1/3 < x < y_1/2$, all factors are nonnegative, and the first is at most $1/2$. Thus $1 - R_y(x) \geq 1/2$ there. Integrating the envelopes, and splitting the upper integral at $1/S_y$, yields

$$b_p \mathbb{E}_w y_1^{-p} \leq J_N(w) \leq B_p \mathbb{E}_w S_y^p, \tag{29}$$

where one may take $b_p = a_*(3^p - 2^p)/(2p)$ and $B_p = 2A((1-p)^{-1} + p^{-1})$. These are uniform in N. The inequality survives sign changes of individual factors; replacing the product by a positive product on the entire integration interval would not be valid.

8.2 Association and the reference law

For $K(x, y) = (y - x)^2/(x + y)$ on $x < y$,

$$\partial_x \partial_y \log K(x, y) = 2/(y - x)^2 + 1/(x + y)^2 > 0.$$

An ordered density proportional to $\prod_i w_i(x_i) \prod_{i<j} K(x_i, x_j)$ is positively associated. One direct proof conditions on the largest coordinate. By induction the remaining law on the shorter interval is associated. The ratio of its density at a smaller largest coordinate to that at a larger one is decreasing in all remaining coordinates, including the support indicator. It therefore lowers increasing expectations. Total covariance now has a nonnegative conditional part and a nonnegative covariance of monotone conditional expectations. Truncation handles unbounded integrable observables. Hence an increasing likelihood ratio moves all coordinates upwards in stochastic order, and a decreasing ratio moves them downwards. This is the MTP_2 mechanism used in the source, not a new association principle.

Let $\mathbb{E}_{n,a}$ denote the ordered Bures–Laguerre density

$$Z_n(a)^{-1} \prod_i r_i^{a-1} e^{-r_i} \prod_{i<j} K(r_i, r_j), \quad 0 < r_1 < \cdots < r_n < \infty, \quad a > 0. \tag{30}$$

Its ordered normalizer is

$$Z_n(a) = \prod_{i=0}^{n-1} \Gamma(a + i) \prod_{0 \leq i < j < n} \frac{j - i}{2a + i + j}, \qquad Z_0(a) = 1. \tag{31}$$

To verify it, write the pair product as a Vandermonde determinant times the Schur Pfaffian $\mathrm{Pf}[(r_j - r_i)/(r_j + r_i)]$, using a border of ones in odd size. The double moment is

$$\iint r^{a+i-1} s^{a+j-1} e^{-r-s} \frac{s - r}{s + r} \, \mathrm{d}r \, \mathrm{d}s = \Gamma(a + i)\Gamma(a + j) \frac{j - i}{2a + i + j}.$$

The change to $r + s$ and $r/(r + s)$ evaluates it. The finite determinant–Pfaffian integration identity and Schur identity then give (31); absolute integrability follows from the gamma moments and $|(s - r)/(s + r)| \leq 1$. The ordered normalization is important: an extra $n!$ would spoil the ratios below.

Lemma 8.3 (Reference estimates for any $0 < p < 1/2$)**.** *Uniformly in $n \geq 1$,*

$$\mathbb{E}_{n,1/2}\left(\sum_i r_i^{-1}\right)^p \leq C_p n^{2p}, \tag{32}$$

$$\mathbb{E}_{n,1} r_1^{-p} \geq c_p n^{2p}. \tag{33}$$

Also $\Pr_{n,1/2}(r_n \leq Kn)$ is bounded below uniformly in n for a fixed sufficiently large K.

Proof. For general fixed $a > 0$ and $D_k = ak + k(k-1)/2$, integrating the first k coordinates below h, bounding their pair factors by h and cross factors by the larger coordinate, gives

$$\Pr_{n,a}(r_k < h) \leq \frac{h^{D_k}}{a^k k!} \frac{Z_{n-k}(a+k)}{Z_n(a)}.$$

The exact ratio from (31) is

$$\prod_{i=0}^{k-1} \Gamma(a+i)^{-1} \prod_{\substack{0 \leq i < j < n \\ i < k}} \left(1 + \frac{2a+2i}{j-i}\right).$$

Gaps at most k contribute at most $e^{O_a(k^2)}$ by summing $\log(1 + C_a k/j)$. Larger gaps contribute at most $(n/k)^{2D_k}$. Finally, $\prod_{i<k} \Gamma(a+i) \geq k^{D_k} e^{-O_a(D_k)}$: integrate each gamma integral over $[a+i, a+i+1]$, sum logarithms, and use $x \log(x/k) \geq -k/e$. Terms $k \log k$ and k are absorbed in $O_a(D_k)$, with bounded k handled separately. Therefore

$$\Pr_{n,a}(r_k < h) \leq \min\left(1, \left(\frac{C_a h n^2}{k^3}\right)^{D_k}\right). \tag{34}$$

Integration of this tail gives, for $0 < q < D_k$,

$$\mathbb{E}_{n,a} r_k^{-q} \leq \frac{D_k}{D_k - q} (C_a n^2/k^3)^q. \tag{35}$$

At $a = 1/2$, use $q = p$ for $k = 1$ and $q = 1$ for $k \geq 2$. Then $\mathbb{E} r_1^{-p} \lesssim n^{2p}$ and $\mathbb{E} \sum_{k \geq 2} r_k^{-1} \lesssim n^2$. Subadditivity of the p-power and concavity give (32). At $a = 2$, use $q = 1$ for all k to get $\mathbb{E}_{n,2} \sum r_i^{-1} \lesssim n^2$.

For the lower estimate, the density of $r_1 = s$ at $a = 1$ equals

$$e^{-s} \frac{Z_{n-1}(2)}{Z_n(1)} \mathbb{E}_{n-1,2} \prod_i \left[\mathbf{1}_{\{r_i > s\}} \frac{(1 - s/r_i)^2}{1 + s/r_i}\right].$$

The ratio is $n(n+1)/2$. Each bracket is in $[0,1]$, and its deficit is at most $3s/r_i$. The product is at least $1 - 3s \sum_i r_i^{-1}$. For $s \leq \varepsilon n^{-2}$, choose a fixed small ε so the expectation is at least $1/2$. The density is then bounded below by a positive multiple of n^2. Integrating over a fixed subinterval of $(0, \varepsilon n^{-2})$ proves (33). The case $n = 1$ follows either from the same empty-product convention or from its exponential density.

Finally $K(r,s) \leq s$ for $r < s$ gives

$$\Pr_{n,a}(r_n > Kn) \leq \frac{Z_{n-1}(a)}{Z_n(a)} \int_{Kn}^{\infty} s^{a+n-2} e^{-s} \, \mathrm{d}s.$$

The ratio is at most $e^a (2e)^{n-1}/\Gamma(a+n-1)$, and the normalized gamma tail is at most $2^{a+n-1} e^{-Kn/2}$. A sufficiently large fixed K makes the probability uniformly less than $1/2$. This completes the proof. □

8.3 Proof of the extension

Proof of Theorem 8.2. For the upper bound, first condition the target ensemble on $y_N < 1/2$. This decreasing event increases every decreasing inverse statistic, by association. Compare on that interval with $y_i = r_i/(bN)$ in (30) at $a = 1/2$, conditioned on $r_N < bN/2$. The target-to-reference likelihood ratio is proportional to $\prod_i w(y_i) y_i^{1/2} e^{bNy_i}$. Its logarithmic derivatives are nonnegative if $b \geq D$. Choosing also $b/2 \geq K$ makes the reference conditioning probability uniformly bounded below. Consequently

$$\mathbb{E}_w S_y^p \leq C(bN)^p \mathbb{E}_{N,1/2}(\sum_i r_i^{-1})^p \leq C_w N^{3p}.$$

Together with (29) this proves the upper bound.

For the lower bound, change variables $y_i = 2z_i/(1+z_i)$. The transformed pair kernel is

$$K\left(\frac{2z_i}{1+z_i}, \frac{2z_j}{1+z_j}\right) = \frac{2K(z_i, z_j)}{(1+z_i)(1+z_j)(1+2z_iz_j/(z_i+z_j))}.$$

Compare with $z_i = r_i/(bN)$ at $a = 1$, conditioned on $r_N < bN$, where $b = 1/4$. The ratio consists of one-variable factors

$$w\left(\frac{2z}{1+z}\right)(1+z)^{-(N+1)} e^{bNz}$$

and decreasing pair factors $(1 + 2z_iz_j/(z_i+z_j))^{-1}$. The logarithmic derivative of each individual factor is at most $E - (N+1)/(1+z) + N/4 \leq E - N/4 - 1/2$. Thus for every sufficiently large N the entire ratio decreases coordinatewise. Applying association to the *reference* law puts the target z coordinates below that conditional law, itself below the unconditioned reference. A lower bound on the conditioning probability is not needed for this direction. Since $y_1 \leq 2z_1$,

$$\mathbb{E}_w y_1^{-p} \geq 2^{-p}(bN)^p \mathbb{E}_{N,1} r_1^{-p} \geq c_w N^{3p}.$$

Equation (29) proves the desired lower bound for large N. Each of the finitely many smaller N has finite positive J_N by Theorem 4.1 and the pointwise lower bound, so reducing the constant covers them all. □

8.4 Two concrete classes and a sharp integrability boundary

The manuscript's weight is admissible with $p = 1/4$. It is decreasing, and

$$(\log(\omega(x)x^{1/2}))' = \frac{1 - 2x - \sqrt{2x/(1+x)}}{4x(1-x)} \geq -4 \quad (0 < x \leq 1/2).$$

For $x \leq 1/8$ its numerator is nonnegative; on the remaining interval the elementary denominator and numerator bounds give the displayed constant. The upper and lower power envelopes follow from $x^{1/4}\omega(x) = \frac{1}{2}\sqrt{\sqrt{(1+x)/2} - \sqrt{x}}$, a positive continuous function on $[0, 1/2]$. If $V \in C^1(0,1)$ is bounded with $|V'| \leq L$, then $w = e^V\omega$ has $D = 4 + L$ and $E = 2L$, with appropriately scaled power envelopes. Thus

$$J_N(e^V\omega) \asymp N^{3/4} \tag{36}$$

with constants independent of N. Unlike (24), this conclusion does not lose an exponential factor with height.

Likewise, for $0 < p < 1/2$, $w = x^{-p}e^V$ is admissible with $D = L, E = 2L$. It gives the full family $J_N(w) \asymp N^{3p}$. This is an extension of the positive-integral analysis, not a claim of new SAW critical exponents or a historical-priority assertion for Bures ensembles.

Proposition 8.4 (Why $p < 1/2$ cannot simply be dropped)**.** *For $w(x) = x^{-p}$, $1/2 \le p < 1$, the ensemble still has a finite positive normalizer, but $J_N(w) = +\infty$ for every $N \ge 1$.*

Proof. The normalizer is finite since $p < 1$ and the pair kernels are bounded. For $N \ge 2$, restrict $y_2, \dots, y_N$ to disjoint compact intervals inside $(1/2, 1)$ and let y_1 tend to zero. All pair factors involving these separated coordinates are bounded below by positive constants; the marginal integrand for y_1 is therefore bounded below by a constant times y_1^{-p}. The pointwise lower-integral bound is a positive constant times y_1^{-p}, so its expectation contains $\int_0^{\varepsilon} y_1^{-2p}\,\mathrm{d}y_1 = \infty$. For $N = 1$ the same argument uses its one-variable density directly. □

The divergence is logarithmic at the endpoint. It is not cured by choosing more moment coordinates or by an exact finite inverse of M.

9 From moments to bridge bounds: the remaining gates

Proposition 9.1 (Quantitative monotone increment lemma)**.** *Suppose $B_N \ge 0$ decreases, $0 < \alpha < 1$, and for all $N \ge 1$,*

$$aN^{\alpha} \le m_N \le bN^{\alpha}, \qquad uB_N \le m_{N+1} - m_N \le vB_N,$$

where $a, b, u, v > 0$. For $N \ge 2$,

$$\frac{aL^{\alpha} - b}{v(L-1)} N^{\alpha-1} \le B_N \le \frac{2b}{u} N^{\alpha-1}, \tag{37}$$

whenever the fixed integer $L > 1$ satisfies $aL^{\alpha} > b$.

Proof. Summing the lower increment estimate for $j = 1, \dots, N-1$ gives $u(N-1)B_N \le m_N - m_1 \le bN^{\alpha}$. For the other direction, $(aL^{\alpha} - b)N^{\alpha} \le m_{LN} - m_N \le v\sum_{j=N}^{LN-1} B_j \le v(L-1)NB_N$. □

At $p = 1/4$, the source identification and its monotonicity/increment lemma supply these gates with $\alpha = 3/4$. Ram coordinates preserve this deduction but do not remove its hypotheses. In the general weight theorem, $3p$ may even exceed one. It would then be inconsistent to infer a decreasing bounded bridge mass of power $3p - 1$ without changing the model. No such inference is made here.

For finite computations at N and LN, interval bounds for the two moments can furnish a lower bound for B_N *if* a valid global upper increment constant v is independently known. Conversely, a sequence of finite interval evaluations with no uniform tail or increment control is not an all-height theorem.

10 Diagnostics and reproducibility

The companion evidence script uses rational atomic measures to test the exact response system, the rank-one identity, Cauchy determinant positivity, the polynomial Ram formula, and the observable identity against direct enumeration of every ordered subset in the finite fixtures. Its executed run passed 187 exact finite checks with zero failures. It separately tests the adjoint correction and algebraic normalizer ratios. Atomic fixtures are not the continuous weight ω; they verify finite identities and implementation conventions, not the source's continuum asymptotic.

The endpoint and hard-edge theorems are justified by the proofs above. Their all-height quantifiers are not inferred from plots, finite samples, or a fitted slope. In particular, this report

does not present an interval-certified large-N table for the physical strip mass, a benchmark showing faster evaluation than structured Cauchy algorithms, or a completed Lean formalization.

The local evidence package records executed checks, source hashes, actual read scopes, the source-stage review, and PDF-generation diagnostics. Only two minor-formatting review rounds are allowed; scientific hypotheses are not relaxed merely to achieve a successful typeset build. Compiler success is not evidence for mathematical validity.

11 Advantages, limitations, and research directions

The useful Ram contribution is to expose exactly which finite response is inverted and which physical functional is evaluated. Localization and finite moments give an exact representation on the selected polynomial space. The Laplace realization provides a positive physical metric while preserving the derivative coupling and boundary term. Primal–dual bounds separate coefficient residuals, singular endpoint errors, interior quadrature errors, and unresolved infinite tails. The weight-sensitivity calculation keeps normalization and both weight arguments; the hard-edge extension replaces exponentially weak generic perturbation estimates by a uniform power law under explicit slope assumptions.

The costs are equally concrete. Cauchy bimoment systems can be badly conditioned as N grows, and this report proves no uniform inverse bound. Gram whitening changes arithmetic, not the underlying finite approximation space or its mathematical information. Endpoint enclosures may dominate the error budget. The general integral extension does not construct a local lattice deformation, preserve Yang–Baxter identities, or prove a new path theorem. The original transfer and Pfaffian machinery has structural information not supplied by a generic moment bank. Classical biorthogonal and Pfaffian algorithms may be more efficient; that comparison requires measurements not supplied here.

Promising next steps are to obtain certified lower singular-value bounds in an appropriate physical scaling, exploit the rank-one displacement structure without forming normal equations, and combine interval quadrature with the explicit endpoint and Laplace-tail estimates. A second direction is to determine which perturbations $e^{V}\omega$ actually arise from a positive local walk model whose flux and increment identities survive. A third is to replace the sufficient logarithmic-slope conditions by weaker order or transport conditions, while retaining the sharp integrability boundary. Finally, a complete independent audit or formal proof of the source's transfer-to-polynomial steps would be needed before claiming an independently verified new global lattice result.

12 Contribution assessment and conclusion

Applicability remains L4: the source contains a directly usable Cauchy moment problem, but Ram methods are not a substitute for its entire integrable-model proof. Contribution is assessed C3/5 for the explicit weight-class extension, normalization-aware sensitivities, and the source-specific endpoint/dual certification route. The transform identity, positive determinant argument, association mechanism, gamma normalizer, and standard adjoint residual principle are credited to their classical origins rather than counted as inventions. No historical-priority claim, better honeycomb exponent, conformal limit, global proof certification, or measured speedup is included in this score.

The main constructive conclusion is precise: the positive integral admits a faithful finite Ram representation, and its hard-edge power persists throughout an explicitly controlled class of weight changes. The same analysis identifies why symmetric-Gram substitution, uncertified finite residuals, and unrestricted endpoint perturbations would fail. These results improve the transparency and

extendibility of a specific proof stage while leaving the remaining geometric and uniformity obligations visible.

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